



Fig. 3: iDose⁴ images of a patient who had undergone coronary angiography and stenting. iDose⁴ improved spatial resolution enabled clear visualisation of the patency of the stent

When this very high level of noise is propagated through the reconstruction algorithm, the result is an image with significant artefacts and high-quantum mottle noise."

iDose⁴ is based on a completely different approach to image reconstruction (IR) techniques. Raveendran Gandhi says, "IR techniques attempt to formulate image reconstruction as an optimisation problem, i.e., IR attempts to find the image that is the 'best fit' to the acquired data. The noisiest measurements are given low weight in the iterative process; therefore they contribute very little to the final images." He further adds, "Hence, IR techniques treat noise properly at very low signal levels, and consequently, reduce the noise and artefacts present in the reconstructed image. This results in an overall improvement of image quality at any given dose." With IR techniques, the noise can be controlled for high spatial resolution reconstructions, hence, providing high-quality, low-contrast and spatial resolution within the same image. These are possible due

to the recent innovations in hardware design and algorithm optimisations.

Working of iDose⁴. Whilst explaining the working of iDose⁴, a noteworthy point mentioned by Raveendran Gandhi is that in iDose⁴ algorithm, iterative processing is performed in both the projection and image domains. The reconstruction algorithm starts with projection data, where it identifies and corrects the noisiest CT measurements—those with very poor signal-to-noise (S/N) ratio, or very low photon counts. Each projection is examined for points that have likely resulted from very noisy measurements using a model that includes the true photons statistics. Through an iterative diffusion process, the noisy data is penalised and edges preserved. This process ensures that the gradients of underlying structures are retained, thus, preserving spatial resolution whilst allowing a significant noise reduction.

The noise that remains after this stage of algorithm is propagated to the image space. However, the propagated noise is now highly localised and can

be effectively removed to support desired level of dose reduction. Then the iDose⁴ algorithm deals with subtraction of the image noise whilst preserving the underlying edges associated with true anatomy or pathology.

Challenges faced. With a healthy innovation comes obstacles that need to be tackled. Talking about the challenges the Philips team faced whilst coming up with iDose⁴, Raveendran Gandhi says, "While transitioning from the existing, widely accepted FBP technology to IR technology through our basic first-generation iDose platform, and its innovative journey to graduate into the current fourth generation of iDose⁴, optimising and improving the product in every stage, we have surpassed several challenges." Radiologists appreciate images, which have a natural appearance, devoid of all artefacts. "As each CT patient study may contain acquisitions of several hundred cross-sectional images of the body, those images were expected to be reconstructed under a minute to match with the busy patient workflow of a radiology department. Such ultrafast reconstruction requires very high computational power. Philips achieved reconstruction speed of 20 images per second through its Rapid-view IR hardware. The other biggest challenge was to improvise the existing methods to significantly improve spatial resolution in order to drastically improve image quality. Despite these

The primary objective of iDose⁴ is to address the radiation dose issues and provide the real benefit of low-dose, high-quality images to patients and radiologists. The focus is to create and spread awareness about low-dose technology available